

GAYATHRI H

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LeetCode : <https://leetcode.com/u/GayathriHGayathriH/>



CAREER OBJECTIVE

Aspiring Software Engineer and Artificial Intelligence and Data Science student with skills in Python, ML, OOPS, DSA and SQL, offering expertise in data analysis, database management, and problem-solving. Seeking an opportunity to contribute to business insights while pursuing professional growth and continuous learning.

EDUCATION QUALIFICATION

- **Bachelors of Engineering-Artificial Intelligence & Data Science**
CMR Institute of Technology, Bengaluru
CGPA: 9.2, 2027(Pursuing)
 - **Pre-University Course- Science**
SFS PU College, Kolar
94%, 2023
 - **SSLC**
SFS High School, Malur
92%, 2021
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TECHNICAL SKILLS

Programming: C, Python, OOPS Concepts.

Operating System: Windows, Ubuntu, Linux.

Database: SQL, MongoDB

Data & Visualization Tools: NumPy, Power BI

Cloud Computing: Cloud Fundamentals, Virtual Machines, Containers.

DevOps: Docker, Kubernetes (Basics), CI/CD Concepts.

Web Technologies: HTML, CSS, JavaScript(Basics), React.js, REST API's.

Tools: Git, GitHub, VS Code, Jupyter Notebook

PROJECTS

- **Eco-barrier – AI Based Smart Elephant Deterrent System:**
 - Developed a smart wildlife intrusion detection system using Arduino, PIR sensor, and ultrasonic sensor for real-time elephant detection at farm boundaries.
 - Implemented a two-stage verification pipeline where Arduino handles initial motion sensing and Raspberry Pi runs a YOLO-based AI model to confirm elephant presence with confidence thresholding.
 - Designed a non-harmful acoustic deterrent system using a relay-controlled directional speaker emitting bee-buzz frequency sounds (3–10 kHz) that activates only upon AI-verified detection.
 - Tech Stack: **Python, OpenCV, YOLOv8, Arduino Uno, PIR Sensor, Pi Camera Module V2, GSM SIM800L, Relay Module.**
- **Self-Correcting RAG System**
 - Developed a Retrieval-Augmented Generation (RAG) system to answer questions from PDFs, scanned images, and DOCX files using OCR, semantic search, and LLMs.
 - Implemented a self-correction layer with confidence scoring, query reformulation, contradiction detection, and low-confidence responses to reduce hallucinations.
 - Built an end-to-end pipeline for document ingestion, semantic chunking, embeddings, vector search, reranking, and evaluation.
 - Tech Stack: **Python, LangChain, FAISS/ChromaDB, Sentence Transformers, OpenAI GPT/Llama 3, PyMuPDF, EasyOCR, Streamlit.**
- **VigilNethra – AI-Powered Intelligent Surveillance System**
 - Developed an AI-powered surveillance system for real-time object detection and monitoring using computer vision and deep learning techniques.

- Implemented YOLOv8-based object detection with OpenCV to analyze live video streams, detect suspicious activities, and generate automated alerts through a FastAPI backend.
- Built a full-stack application with a React frontend and RESTful APIs, enabling real-time surveillance, alert management, and seamless communication between frontend and backend services.
- Tech Stack: **Python, FastAPI, React.js, YOLOv8, OpenCV, REST APIs, SQLite/PostgreSQL, Git, GitHub, Docker.**

INTERNSHIP

- **Company name:** Cod Soft
Duration: 1 Month
Designation: Data Science Intern (Virtual Internship)
Responsibilities:
 - Developed machine learning projects including Titanic Survival Prediction, Sales Prediction, and Credit Card Fraud Detection using Python.
 - Analyzed datasets to identify patterns and improve model accuracy through evaluation metrics.
 - Prepared technical documentation and collaborated on team-based project development.
 - Tech Stack: Python, Pandas, NumPy, Scikit-learn, Machine Learning, Data Analysis

CO-CURRICULAR & EXTRA CURRICULAR ACTIVITIES

- **Hackathons:**
 - Participated in Smart India Hackathon organized by Government of India, 2024.
 - Participated in Google Cloud Agentic AI Day powered by H2S.
 - Participated in Adobe India Hackathon.
 - Participated in Myntra Hacker Ramp HeForShe 2025.
- **Seminars/Webinars/Workshops:**
 - Conducted a seminar in promoting the appropriate Technologies on the topic of Menstruation at Penial High school during August 2025.
 - Attended Power BI Workshop conducted by Office Master.
 - Has attended an exclusive one-day offline workshop at the Geeks for Geeks Classroom Centre.
- **NSS:**
 - Took part in awareness rally to promote social causes under NSS.
 - Contributed to NSS activity conducted in government school.
- **Cultural Fests:**
 - Volunteered for CULTURA-2024 at CMRIT by supporting coordination and smooth execution of events in security desk.
 - Volunteered for SKS event at CMRIT.
- **Non-Technical Club Activities:**
 - Volunteered in Google Cloud presents Agentic AI Day Powered by H2S in 2025.
 - Volunteered in youth empowerment and community development programs through the C2C Fellowship - yuvaka sangha in 2025.
- **Additional Responsibilities:**
 - Member of Peer Supporting and Peer Learning Team at CMRIT.
 - Serving as Class Representative for Artificial Intelligence and Data Science department.

AWARDS AND ACHIEVEMENTS

- Awarded Elite + Silver (Top 5%) in NPTEL - "Python for Data Science" with consolidated score of 83% in the proctored examination conducted by IIT, Madras.
- Recognized as Department Topper in 5th Semester at CMRIT for outstanding academic performance.

PERSONAL DETAILS

Date of Birth	: 03-01-2006
Nationality	: Indian
Gender	: Female
Linguistic Competence	: Kannada, English, Hindi, Telugu
Hobbies	: Listening to music, Badminton.